

GAITHERSBURG MONTGOMERY, MD, USA

WMO: 720334

Lat: 39.167N

Lon: 77.167W

Elev: 539

StdP: 14.41

Time Zone: -5.00 (NAE)

Period: 07-19

WBAN: 93764

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 10.1	(c) 15.7	(d) -4.3	(e) 4.5	(f) 12.4	(g) 0.5	(h) 5.8	(i) 17.5	(j) 25.3	(k) 33.5	(l) 23.0	(m) 30.9	(n) 6.1	(o) 300	(p) 0.413

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.4	91.2	75.5	89.9	74.8	87.7	73.9	77.8	88.9	76.5	86.5	75.2	83.9	6.3	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.3	130.9	83.3	73.1	125.6	81.3	72.4	122.3	80.4	41.5	88.4	40.1	87.1	38.9	83.8	83.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.7	16.8	14.1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			3.6	80.2	3.7	1.8	0.9	81.5	-1.3	82.5	-3.4	83.5	-6.1	84.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	55.2	32.0	35.5	43.1	54.4	65.6	72.3	76.3	74.0	68.3	56.8	44.8	37.9
	DBStd	17.39	9.86	10.35	9.96	8.86	7.89	5.68	4.98	4.63	6.74	8.35	8.24	8.89
	HDD50	1911	563	420	263	50	1	0	0	0	0	28	197	389
	HDD65	4742	1023	825	679	337	95	11	0	1	42	283	607	839
	CDD50	3807	5	15	50	182	484	670	814	745	548	237	41	16
	CDD65	1161	0	0	1	18	113	231	349	281	140	27	1	0
	CDH74	9999	0	3	18	249	965	1997	3246	2276	1058	185	2	0
	CDH80	3460	0	0	0	66	272	659	1303	772	349	39	0	0
Wind	WSAvg	4.9	6.2	6.5	6.4	6.1	4.6	4.1	3.5	3.3	3.3	4.4	5.2	5.6
Precipitation	PrecAvg	44.8	3.4	2.6	3.8	3.6	4.3	4.3	4.1	4.0	4.0	3.9	3.0	3.6
	PrecMax	66.2	6.2	6.0	8.0	6.3	8.8	10.7	7.9	8.1	10.8	9.6	6.7	6.6
	PrecMin	33.5	1.7	0.4	0.0	1.9	1.2	0.6	1.2	0.9	0.0	0.0	0.9	1.7
	PrecStd	7.3	1.2	1.4	1.7	1.2	2.0	2.4	1.7	1.7	2.4	2.3	1.5	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	64.1	71.7	76.3	87.7	90.2	92.8	96.9	92.3	91.2	84.1	72.2	66.4
		MCWB	55.8	59.3	59.7	64.6	74.2	76.3	77.8	76.2	75.1	72.5	60.8	61.6
	2%	DB	57.1	63.9	71.6	81.3	86.5	90.0	92.0	90.1	88.2	79.4	66.4	62.7
		MCWB	51.8	54.0	58.4	63.4	71.8	74.8	76.1	75.4	73.5	68.8	58.0	58.5
	5%	DB	52.2	57.0	65.5	75.2	82.3	87.6	90.1	87.6	83.9	74.6	63.1	56.5
		MCWB	46.4	48.7	54.8	60.6	69.3	73.8	75.4	74.4	72.4	65.9	54.9	51.5
	10%	DB	46.0	51.8	60.8	71.6	79.9	83.8	87.7	84.1	80.7	71.2	58.6	51.8
		MCWB	41.2	44.5	50.4	59.0	67.6	72.0	74.7	72.5	69.3	64.2	51.6	47.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	58.3	61.4	62.5	68.4	75.6	77.8	80.2	78.6	77.5	74.4	63.9	62.8
		MCDB	62.3	68.3	72.2	77.5	86.5	89.7	92.6	89.2	87.6	80.9	69.4	64.8
	2%	WB	53.1	56.0	59.6	65.5	73.6	76.2	78.2	77.1	75.4	71.3	59.7	58.8
		MCDB	57.2	62.3	68.3	77.5	83.2	87.2	89.8	87.0	84.1	76.3	64.5	62.8
	5%	WB	46.5	49.8	55.9	62.7	71.4	74.6	76.9	75.4	73.7	68.6	56.6	53.1
		MCDB	50.2	56.6	65.0	73.9	80.3	84.1	87.3	83.6	81.1	73.4	61.6	56.2
	10%	WB	41.7	44.9	51.6	59.8	69.3	73.2	75.4	74.1	71.7	64.7	53.3	46.9
		MCDB	46.3	50.7	58.8	69.2	77.1	81.3	84.0	81.5	77.4	69.5	57.0	49.7
Mean Daily Temperature Range	5% DB	MDBR	15.4	16.8	18.6	21.4	19.7	19.5	19.4	18.9	18.6	19.0	17.9	14.8
		MCDBR	22.8	25.2	27.5	27.7	23.3	22.5	22.0	21.6	22.4	23.3	23.2	19.8
		MCWBR	18.5	18.3	17.2	14.6	11.4	10.0	8.8	9.1	10.4	13.1	15.5	17.1
	5% WB	MCDBR	20.9	22.5	24.2	24.2	19.7	20.1	19.2	18.1	19.1	17.9	18.8	18.4
		MCWBR	18.0	18.2	16.7	14.3	10.3	9.9	8.6	8.7	9.6	11.2	14.5	17.3
Clear-Sky Solar Irradiance	taub	0.304	0.318	0.352	0.399	0.440	0.499	0.516	0.514	0.431	0.358	0.319	0.305	
	taud	2.472	2.412	2.390	2.276	2.191	2.064	2.036	2.023	2.232	2.441	2.510	2.504	
	Ebn at Noon	277	287	287	278	268	251	246	242	257	269	269	267	
	Edh at Noon	25	30	35	41	46	52	53	53	40	30	24	23	
All-Sky Solar Radiation	RadAvg	639	912	1210	1542	1730	1915	1886	1678	1369	1017	733	561	
	RadStd	55	104	82	121	192	146	101	89	167	113	62	56	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (72 neighbors)	+0.72	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+211	+119

Nomenclature: See separate page